THE RUBBER FORMULARY

by

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Plastics Design Library

NEOPRENE 207
AIR CONDITIONER HOSE 208
DUROMETER 65 COMPOUND FOR HOSE 209
DUROMETER 63 COMPOUND FOR HOSE 210
DUROMETER 80 COMPOUND FOR HOSE, BRASS ADHESION 212
DUROMETER 70 COMPOUND FOR HOSE, SURFACE GLOSS 213
NON-BLACK HOSE COVER 214
DUROMETER 75 COMPOUND FOR BELTS 215
V-BELT RECOMMENDATION 216
NEOPRENE SEAL 217
DUROMETER 40 COMPOUND FOR ROLLS & GASKETS 218
NEOPRENE ROLL COMPOUND 219
AIR SPRING 220
STARTING POINT FORMULATION FOR WET SUIT 221
EXTRUSION COMPOUND, MICROWAVE CURE 222
EXTRUSION COMPOUNDS 223
FORMULATIONS FOR MILITARY APPLICATION 224
FLAME RESISTANT BRIDGE BEARING PAD 225
HIGH HARDNESS COMPOUND 226
DUROMETER 67 PLY SKIM FOR RFL FABRIC AND CORD 227
DUROMETER 60 COMPOUND FOR PLY ADHESION 228
DUROMETER 53 COMPOUND FOR PLY ADHESION 229
DUROMETER 70, TEXTILE ADHESION & MILL RELEASE 230
DUROMETER 70 COMPOUND FOR LOW SET 231
DUROMETER 70, SEMI-CONDUCTIVE/DIRECT BONDING 232
DUROMETER 80 COMPOUND FOR ADHESION 234
DUROMETER 60, TEAR STRENGTH & HEAT RESISTANCE 236
DUROMETER 70, TEAR STRENGTH & HEAT RESISTANCE 237
DUROMETER 65 COMPOUND FOR ABRASION RESISTANCE 238
PRESS MOLDED CLOSED CELL SHEET SPONGE 239
CELLULAR SHOE SOLING 240

IV.

NEOPRENE

AIR CONDITIONER HOSE

DuPont Dow

Neoprene WRT	100
SBR 1502	5
Maglite D	9
Octamine	1
PE 617A	3
HiSil 233	20
N762	50
Atomite Whiting	20
EPON 828	8
Zinc Oxide	2
Calcium Oxide	4
END-75	1
Total	**223**
Physical Properties – Original	
Press Cured: 45 minutes @ 160°C	
T-B, psi	982
E-B, %	495
Shore A, points	76
Compression Set, Method B	
22 Hours @ 100°C	88

DUROMETER 65 COMPOUND FOR HOSE
PPG

		Banbury 1
Neoprene WRT		100.0
BR 1203		3.0
Magnesia		3.0
Hi-Sil 210		30.0
FEF N550		30.0
ODPA		2.0
LP Oil		10.0
Stearic Acid		2.0
		Banbury 2
Zinc Oxide		5.0
ETU 75%		0.75
Total		185.75
Cure Rate 154°C, T90		20 minutes
Mooney Scorch 121°C, Minutes:		
T5		14
T35		26
Viscosity, ML 100		60
Cure: 40'/154°C		
Durometer		66
M20, MPa (PSI)		1.0(144)
M300		9.1(1330)
Tensile		14.4(2100)
Elongation		415%
Aging at 100°C	3 Days	14 Days
Durometer	73	78
Tensile, MPa (PSI)	14.3(2100)	14.0(2050)
Elongation %	375	335
Percent E_B Retained	90	81
Cure: 50'/154°C		
Pendulum Rebound (Z)		
At 23°C		38
At 100°C		61
Compression Set: 3 Days 100°C		48%
DeMattia Cut Growth 23°C		
500% Growth		30 kilocycles

DUROMETER 63 COMPOUND FOR HOSE

PPG

				Banbury 1
Neoprene				100.0
BR 1220				5.0
Hi-Sil 532EP				50.0
Hi-Sil 243LD				20.0
Naphthenic Oil				15.0
ODPA				2.0
				Banbury 2
Zinc Oxide				5.0
Magnesium Oxide				2.0
Stearic Acid				2.0
Armoslip CP				1.0
Specific Gravity				1.44
Neoprene Type			GRT	WRT
PEG 3350			--	3
TMETD			--	1
HMT			--	2
Cure Rate MDR 154C, T50			8.2	8.1
T90			23	21
Mooney Scorch 121C, T5			30+	12
Mooney Viscosity, ML100			32	47
Cured 20'/30' @ 154C				
Durometer: at 23C			63	62
at 100C			56	55
Stress/Strain	Original	Aged*	Original	Aged
Durometer	63	80	62	74
M300, MPa	2.7	--	2.4	--
Tensile	13	11	10	9
Elongation, %	920	585	810	555
*Aged 72 hours @ 100C				
Immersions, 72 hrs.	ASTM #1 Oil 100C	Water 60C	ASTM #1 Oil 100C	Water 60C
% Volume Increase	-0.8	7.3	1.0	11.1
Durometer	58	58	55	55
Tensile	12	12	10	11
Elongation, %	730	815	755	780

DUROMETER 63 COMPOUND FOR HOSE
Continued

MG Trouser Tear, kN/m	29	29
Die C Tear, kN/m	42(240)	29(165)
Compression Set, 70 hrs. @ 100C, %	96	82
Pendulum Rebound (Z), %		
At 23C	42	42
At 100C	51	51
DeMattia Cut Growth, KC to 500%	30	13
Kinetic Coefficient of Friction		
Wet Aluminum	1.06	1.23

DUROMETER 80 COMPOUND FOR HOSE, BRASS ADHESION

PPG

Neoprene W	100.0
Magnesium Oxide	4.0
Silene D	50.0
Hard Clay	20.0
GPF N660	40.0
Naphthenic Oil	18.0
Microcrystalline Wax	2.0
Stearic Acid	2.0
Resorcinol Resin	2.0
Wingstay 100	1.5
Zinc Oxide	5.0
DOTG	1.0
TMTM	1.0
Sulfur	1.0
Cyrez 936P	5.0
Total	252.5
Cure Rate @ 154°C, 90% ODR	15’
Mooney Scorch @ 121°C, T5	26’
Viscosity: ML 100	88
ODR Minimum	7.4
Stress-Strain Properties (Cure: 30’/154°C)	
Hardness	79
M100	710(4.9)
Tensile	1640(11.2)
Elongation	290
Compression Set, 70 hr @ 100°C	76%
Brass Strip Adhesion (lbs./in), 25’/154°C	30

DUROMETER 70 COMPOUND FOR HOSE, SURFACE GLOSS
PPG

Neoprene WHV	50.0
Neoprene W	25.0
Neoprene WB	25.0
Magnesium Oxide	3.0
Silene D	75.0
Hard Clay	90.0
Aromatic Oil	30.0
TCP	20.0
Red Oxide	3.0
Stearic Acid	2.0
Zinc Oxide	2.0
Trimethyl Thiourea	1.0
Total	329.0
Cure Rate @ 154°C; 90% ODR	23
Mooney Scorch @ 121°C, T5	19
Viscosity: ML 100	82
ODR Minimum	6.7
Stress-Strain Properties (Cure: 40'/154°C)	
Hardness	68
M300	590(4.0)
Tensile	1010(6.9)
Elongation	600

NON-BLACK HOSE COVER

R.T. Vanderbilt

Neoprene W	100.0
Stearic Acid	0.5
VANOX MBPC	2.0
Magnesium Oxide (High Activity)	4.0
DIXIE CLAY	140.0
Light Process Oil	15.0
Picco Resin 6100	5.0
VANWAX H	2.0
Zinc Oxide	5.0
END-75	1.0
ALTAX	0.75
Total	275.25
Properties, Cured 30 Minutes at 152°C(307°F)	
100% Modulus, MPa (psi)	2.9(425)
300% Modulus, MPa (psi)	10.8(1575)
Tensile Strength, MPa (psi)	12.3(1825)
Elongation, %	360
Hardness, Shore A	62
Tear Strength, Die C kN/m(pli)	45.4(260)
Mooney at 121°C(250°F)	
Scorch, t5, minutes	22
Plasticity, MS	26

DUROMETER 75 COMPOUND FOR BELTS
PPG

		Banbury 1
Neoprene GN		100.0
Magnesia		1.0
MT N990		100.0
Hi-Sil 233		20.0
LP Oil		10.0
Wingstay 100		2.0
ODPA		1.0
MC Wax		2.0
Stearic Acid		2.0
		Banbury 2
Zinc Oxide		5.0
Total		243.0
Specific Gravity		1.45
Cure Rate 165°C: T75		5 minutes
T90		15 minutes
Mooney Scorch 121°C, T5		29 minutes
Viscosity ML 100		77
Stress-Strain Cure 20/165°C	Original	Aged 72 hr. 100°C
Durometer	75	83
M100, MPa (PSI)	4.6(670)	6.8(1000)
Tensile	12.3(1800)	11.6(1700)
Elongation	320%	250%
Dynamic Properties		
DeMattia Flex 23°C, Not Punched		
100 Kilocycles, No Cracking		
Goodrich Flexometer: 100°C; 20%; 1 MPa		
Static Compression		9.4%
Dynamic Compression		0.2%
Set		4.8%
Heat Build-Up		35°C

V-BELT RECOMMENDATION

DuPont Dow

Neoprene GRT	100
N650 Carbon Black	35
Agerite Stalite	2
Maglite D	4
HiSil 233	15
MBTS	0.5
Stearic Acid	1
SRF 1501	2
Zinc Oxide	5
Cyrex 963	6
Total	**170.5**
Physical Properties	
Tensile, psi	2828
Elongation, %	301
Hardness, Shore A	83
Die C Tear @ RT, ppi	**292**
Block Adhesion	
Polyester	64
Fiberglass	40
Kevlar®	53

NEOPRENE SEAL
R.T. Vanderbilt

Neoprene W	100.0
Stearic Acid	1.0
Magnesium Oxide	4.0
AGERITE STALITE S	2.0
SRF Black (N-774)	40.0
Calcium Carbonate	60.0
Dioctyl Sebacate (DOS)	7.5
VANWAX OZ	1.5
Zinc Oxide	5.0
VANAX DOTG	0.7
UNADS	0.75
Spider Sulfur	0.4
THIATE EF-2	0.75
Total	223.65
Physical Properties	
Tensile, MPa (psi)	14.3(2080)
100% Modulus, MPa (psi)	3.1(440)
Elongation, %	310
Hardness, Shore A	66

DUROMETER 40 COMPOUND FOR ROLLS & GASKETS

PPG

Neoprene W		35.0
Neoprene WHV		65.0
Hi-Sil 233		25.0
Plasticizer SC		25.0
Stearic Acid		1.0
ODPA		2.0
Magnesia		4.0
Zinc Oxide		5.0
Ethylene Thiourea		0.5
Total		162.5
Stress –Strain Properties (Cure: 20'/154°C)	Original	Aged 3 D @ 100°C
Hardness	42	46
M300	360(2.5)	430(2.9)
Tensile	2000(13.7)	2100(14.3)
Elongation	820	760
Compression Set		
1 Day @ 70°C		24%
3 Days @ 100°C		44%
Oil Immersion, 3 Days @ 100°C	ASTM #1	ASTM #3
% Volume Change	-9	74
Hardness	46	27
M300	430(2.9)	330(2.2)
Tensile	2170(14.9)	1500(10.3)
Elongation	740	630

NEOPRENE ROLL COMPOUND
DuPont Dow

Neoprene GRT	70
Neoprene WHV	30
Stearic Acid	1.5
Wingstay AZ	2
Octamine	2
Magnesium Oxide	4
N990 Carbon Black	30
Factice	20
Aromatic Process Oil	40
Zinc Oxide	5
AC Polyethyene 1702	2
Total	**206.5**
Physical Properties – Original	
Tensile, psi	1500
Elongation, %	500
Hardness, Shore A	40

AIR SPRING

DuPont Dow

Neoprene GRT	100
Stearic Acid	0.5
Maglite D	4
Naugard PANA	2
Vanwax H Special	3
N990 Carbon Black	15
N550 Carbon Black	15
Sunthene 4240 Naphthenic Oil	10
Tenex 3901*	5
Zinc Oxide	5
Total PHR's	**159.5**
Original Physical Properties	
Tensile Strength @ Break, psi	2950
Elongation @ Break, %	960
Shore A Hardness, Points	49

* Wood Rosin

STARTING POINT FORMULATION FOR WET SUIT
DuPont Dow

Neoprene GW	100
Maglite D	2
Octamine	2
Stearic Acid	0.5
N772 Carbon Black	15
Circosol 4240	15
Calcium Carbonate	25
Brown Factice	25
CBS	0.5
Zinc Oxide	5
ETU	0.5
Azodicaronamide Blowing Agent	10
Total	**220.5**
Density, g/cc	0.25
Cure (1st Stage)	8 min. @ 160°C
Cure (2nd Stage)	15 min. @ 170°C

EXTRUSION COMPOUND, MICROWAVE CURE

R.T. Vanderbilt

ASTM D2000	BE 620
Neoprene TW 100	100.0
Maglite D	2.0
Stearic Acid	1.0
SRF Black (N-774)	65.0
Sundex 790	20.0
VANOX 2-AZ	2.0
VANWAX H Special	2.0
Zinc Oxide	5.0
END-75	2.0
Desical P	5.0
Total	204.0
Cured 20 minutes at 160°C(320°F)	
Tensile Strength, MPa (psi)	15.2(2200)
Elongation, %	300
Hardness, Shore A	65
Heat Aged 70 Hrs. at 100°C(212°F)	
Tensile Strength, % Change	-2
Elongation, % Change	-9
Hardness, Points Change	+4
Compression Set after 22 Hrs. at 100°C(212°F)	
% Set	12
ASTM #3 Oil Aged 70 Hrs. at 100°C(212°F)	
Tensile Strength, % Change	-35
Elongation, % Change	-40
Hardness, Points Change	-17
Volume Change, %	+40

EXTRUSION COMPOUNDS
R.T. Vanderbilt

ASTM D2000	BC 515	BC 720
Neoprene TW	100.0	100.0
Stearic Acid	1.0	1.0
Maglite D	4.0	4.0
VANFRE AP-2	2.0	2.0
VANPLAST R	5.0	5.0
AGERITE STALITE S	2.0	2.0
NEOPHAX A	35.0	15.0
Sundex 790	10.0	---
FEF Black (N-550)	20.0	30.0
THERMAX (N-990)	40.0	55.0
Zinc Oxide	5.0	5.0
THIATE U	1.5	1.5
Total	225.5	220.5
Cured 30 Minutes at 153°C(307°F)		
200% Modulus, MPa (psi)	4.3(630)	13.3(1930)
Tensile Strength, MPa (psi)	11.7(1700)	16.5(2390)
Elongation, %	400	250
Hardness, Shore A	51	68
Tear Die A, kN/m(pli)	10.6(60)	13.2(75)
Cured 5 Minutes at 171°C(340°F)		
200% Modulus, MPa (psi)	5.4(780)	13.8(2000)
Tensile Strength, MPa (psi)	13.0(1880)	16.5(2400)
Elongation, %	370	240
Hardness, Shore A	53	69
Tear Die A, kN/m(pli)	6.2(35)	12.3(70)
Mooney at 121°C(250°F)		
Scorch, t5, minutes	9	6
Plasticity, ML	37	61
Compression Set after 22 Hrs. at 70°C(158°F)		
% Set	34	16

FORMULATIONS FOR MILITARY APPLICATION

R.T. Vanderbilt

MIL-R-6855, Class II & IV	Grade 50	Grade 60	Grade 70
Neoprene WHV	100.0	100.0	100.0
Maglite D	4.0	4.0	4.0
AGERITE STALITE S	2.0	2.0	2.0
WINGSTAY 100-AZ	2.0	2.0	2.0
Stearic Acid	0.5	0.5	0.5
VANFRE M	1.0	1.0	1.0
THERMAX N-990 Carbon Black	60.0	85.0	90.0
SRF (N-774) Carbon Black	---	---	20.0
Arizona 208 Plasticizer	15.0	15.0	15.0
Zinc Oxide	5.0	5.0	5.0
Ethylene Thiourea, 75% in binder	0.95	0.95	0.95
Total	190.45	215.45	240.45
Properties, Cured 30 minutes at 153°C(307°F)			
Hardness, Shore A	48	59	68
Tensile, MPa (psi)	12(1750)	12(1750)	10(1450)
Elongation, %	550	450	350
Density, Mg/m^3	1.35	1.39	1.43
Compression Set after 70 hours at 100°C(212°F)			
Method B, % Set	23	22	22
After Oven Aging 70 hours at 100°C(212°F)			
Hardness, Points Change	+8	+8	+6
Tensile Change, %	+5	0	0
Elongation Change, %	-13	-16	-16
Weight Change, %	-3	-1	-2
After ASTM #1 Oil Aging 70 hours at 100°C(212°F)			
Hardness, Points Change	0	+3	+3
Tensile Change, %	+3	+8	+10
Volume Change, %	0	-2	-1
After Water Aging 70 hours at 100°C(212°F)			
Hardness, Points Change	+3	+2	+2
Volume Change, %	+6	+5	+4
Resistance to			
Ozone (1 ppm, 20% strain)	No cracks	No cracks	No cracks
Cold Bend at -55°C(-67°F)	No cracks	No cracks	No cracks
Crazing of acrylics (Class IV)	None	None	None

FLAME RESISTANT BRIDGE BEARING PAD
R.T. Vanderbilt

Neoprene WRT	100.0
Magnesium Oxide	4.0
Zinc Oxide	5.0
Stearic Acid	0.5
WINGSTAY 100AZ	2.0
AGERITE STALITE S	2.0
SAF (N-110) Carbon Black	20.0
Hi-Sil 233	15.0
Dioctyl Sebacate	10.0
Ethylene Thiourea, 75% in binder	1.0
METHYL TUADS	0.4
Total	159.9
Properties, Cured min/°C	20/153
Hardness, Shore A	60
Tensile, MPa (psi)	20(2850)
Elongation, %	500
Density, Mg/m^3	1.37
Tear Strength, Die C, kN/m (pli)	35(200)
Compression Set after 22 Hours	100°C(212°F)
Method B, % Set	28

HIGH HARDNESS COMPOUND

R.T. Vanderbilt

Neoprene WM-1	80.0
Neoprene FB	20.0
Magnesium Oxide	4.0
Stearic Acid	0.5
AGERITE STALITE S	2.0
THERMAX (N-990) Carbon Black	80.0
SRF (N-774) Carbon Black	100.0
Circo RPO Naphthenic Oil	8.0
Zinc Oxide	5.0
VANAX DOTG	1.0
UNADS	1.0
Sulfur	0.5
Total	302.0
Properties, Cured min/°C	30/153
Hardness, Shore A	92
Tensile, MPa (psi)	12(1700)
Elongation, %	120
Density, Mg/m^3	1.55

DUROMETER 67 PLY SKIM FOR RFL FABRIC AND CORD
PPG

	Banbury 1
Neoprene GN	100.0
BR 1203	5.0
Hi-Sil 210	40.0
Hi-Sil 532EP	15.0
N774	10.0
DOS	10.0
Magnesium Oxide	2.0
Stearic Acid	2.0
DPPD 100	2.0
SDPA 29	1.0
	Banbury 2
Zinc Oxide	5.0
Total	192.0
Specific Gravity	1.40
Cure Rate 154°C, T50	6.7 minutes
T90	26 minutes
Mooney Scorch 121°C, t5	30+ minutes
Viscosity, ML 100	71
Cure: 35'/154°C	
Durometer	67
M300, MPa (PSI)	4.4(640)
Tensile	11.6(1700)
Elongation	720%
MG Trouser Tear, kN/m (PPI)	16.2(83)
Compression Set, 3 Days 100°C	69.7%
Glass Transition Tg (E")	-26°C to -39°C

DUROMETER 60 COMPOUND FOR PLY ADHESION

PPG

		Banbury 1
Neoprene WRT		100.0
BR 1203		5.0
Hi-Sil 210		40.0
FEF N550		25.0
DOS		10.0
Magnesium Oxide		2.0
Stearic Acid		2.0
ODPA		2.0
Amide Wax		0.7
		Banbury 2
Zinc Oxide		3.0
Sulfur		0.5
TMTM		1.0
DOTG		1.0
Total		192.2
Specific Gravity		1.46
Silane A189	0	1
Cure Rate 154°C, T50 minutes	14	15
Mooney Scorch 121°C T5 minutes	30+	30+
Viscosity, ML100	634	100
Mill Tack	Good	Fair
Cure: 35'/154°C		
Durometer	58	60
M300 MPa	5.0	10.8
Tensile	17	21
Elongation	680%	535%
Aging 7 Days 121°C		
Durometer	78	78
Tensile, MPa	17	16
Elongation	200%	165%
MG Trouser Tear: kN/m(Lbs./in)	23.3(135)	12.8(73)
Ply Adhesion: 0.5 mm Skim Between 2 RFL Cord Plies		
Static Strip N/13 mm	96	79
Rubber Cover	100%	30%
Tension Fatigue Life:		
Kilocycles/mm Separation	1.9	0.7
Rubber Cover	100%	80%

DUROMETER 53 COMPOUND FOR PLY ADHESION
PPG

		Banbury 1
Neoprene WRT		100.0
BR 1203		5.0
Hi-Sil 210		50.0
DOS		10.0
Magnesium Oxide		2.0
Stearic Acid		2.0
SDPA29		1.0
DPPD		2.0
DEG		2.0
		Banbury 2
Zinc Oxide		5.0
Sulfur		0.5
TMTM		1.0
DOTG		1.0
Total		181.5
Specific Gravity		1.38
Cure Rate 154°C, T50		14 minutes
Mooney Scorch 121°C, T5		25 minutes
Viscosity, ML100		56
Cure: 25'/154°C	Original	Aged 500 Hrs 90°C
Durometer	53	75
M300, MPa (PSI)	2.8(410)	--
Tensile	15(2200)	16(2300)
Elongation	830%	580%
MG Trouser Tear, kn/m (PPI)		19.2(110)
Glass Transition (DMA) °C		
E' Onset		33
E' Finish		22
E" Peak		27
Tan Delta Peak		22

DUROMETER 70, TEXTILE ADHESION & MILL RELEASE

PPG

		Banbury 1
Neoprene GRT		100.0
BR 1203		3.0
Magnesia		3.0
Hi-Sil 210		15.0
Hi-Sil 532EP		20.0
FEF N550		25.0
LP Oil		10.0
Stearic Acid		2.0
ODPA		2.0
R-F Resin		1.0
		Banbury 2
Zinc Oxide		5.0
HMMM 65%		3.0
Total		189.0
Specific Gravity		1.36
Cure Rate 154°C, T70		16 minutes
Mooney Scorch 121°C, T5		15 minutes
T35		30+ minutes
Viscosity, ML100		54
Mill Sticking: 35°C		Slight-Moderate
60°C		Very Slight
Cure: 20'/154°C	Original	Aged 3 Days 121°C
Durometer	60	83
M300, MPa (PSI)	10.0(1460)	--
Tensile	14.7(2150)	15.2(2300)
Elongation	485%	225%
Adhesion to Untreated Nylon Fabric		
Strip Peel, Minimum	Lbs./in	kN/m
25'/154°C	60	10.5
50'/154°C	60	10.5
Fatigue Properties		
DeMattia Cut Growth		
500% Growth		55 kilocycles
Monsanto Fatigue, 100%		32 kilocycles

DUROMETER 70 COMPOUND FOR LOW SET
PPG

		Banbury 1
Neoprene WRT		95.0
BR 1203		5.0
Magnesia		2.0
Ciptane		25.0
Silene 732D		20.0
Hard Clay		60.0
ODPA		1.0
Stearic Acid		2.0
Naphthenic Oil		8.0
Castor Oil Ester		3.0
Polyethylene LMW		3.0
		Banbury 2
Zinc Oxide		4.0
PEG 3350		4.0
ETU		0.4
Total		232.4
Specific Gravity		1.58
Cure Rate 154°C (Slope Max.)		21 minutes
Mooney Scorch 121°C, T5		22 minutes
Mooney Viscosity ML100		67
Cure: 30'/154°C	Original	Aged 72 Hrs. 121°C
Durometer	70	79
M300, MPa (PSI)	6.6(960)	
Tensile	9.6(1400)	8.6(1250)
Elongation,%	530	260
Compression Set	72 Hrs. 100°C	22 Hrs.70°C
Percent	26	10
DeMattia Cut Growth, KC to 500%, at 23°C		75
At 100°C		35
Water Immersion 100°C, % Volume Increase		
1 Day		11
7 Days		21
14 Days		16
Garvey Extrusion, Edge-Surface		10A

DUROMETER 70, SEMI-CONDUCTIVE/DIRECT BONDING

PPG

Neoprene GRT			80.0
Hydrin T			40.0
BR 1220			5.0
Hi-Sil 243LD			15.0
N351			25.0
ODPA			2.0
Stearic Acid			2.0
Naphthenic Oil			5.0
R-F Resin			1.0
HMMM 65%			3.0
Zinc Oxide			5.0
Magnesia			4.0
MBTS			1.0
Specific Gravity			1.40
Cure Rate, MDR 155C, T50 minutes			5.7
Mooney Scorch, 121C, T5 minutes			15
Mooney Viscosity, ML100			56
Surface Resistivity (D257), ohms/sq.			4×10^6
Static Decay Time (ML-B-81705B)			<0.01 seconds
Strip Adhesion to RFL Nylon, N/25 mm			250
Cure at 155C: 15 & 25 min.	Original	70 hrs. 100C	700 hrs. 90C
Durometer at 23C	69	78	82
at 100C	63		
M300, MPa (psi)	8.1(1200)		
Tensile	16	20	20
Elongation, %	615	460	325
Compression set, %, 70 hrs. 100C			86
Pendulum Rebound, % at 23C			36
at 100C			55
DeMattia Cut Growth, 500%			100+ KC

DUROMETER 70, SEMI-CONDUCTIVE/DIRECT BONDING CONTINUED

Low Temperature Properties, DNA, 1 Hz:	
Dynamic Modulus, E', MPa	
-30C	2100
-20C	830
-10C	155
0C	65
10C	45
20C	35
Glass Transition (E" max.), T_g	-24°C

DUROMETER 80 COMPOUND FOR ADHESION

PPG

		Banbury 1
Neoprene WRT		94.0
BR 1203		3.0
HSR 60%		16.0
Hi-Sil 532 EP		35.0
Hi-Sil 210		20.0
SRF N762		50.0
Plasticizer SC		5.0
DOS		10.0
Stearic Acid		2.0
DPPD 100		2.0
SDPA 29		1.0
Magnesium Oxide		2.0
		Banbury 2
Zinc Oxide		5.0
Sulfur		0.5
TMTM		1.0
DOTG		2.5
Total		249.0
Specific Gravity		1.41
Cure Rate 154°C, T70		20 minutes
Mooney Scorch 121°C, T5		30+ minutes
Viscosity, ML 100		59
Cure: 30'/154°C	Original	Aged 7 Days 100°C
Durometer	78	88
M300, MPa (PSI)	7.9(1150)	--
Tensile	11.0(1600)	11.4(1670)
Elongation, %	440	200
MG Trouser Tear		11 kNm
Cure: 40'/154°C		
Compression Set 3 Days 100°C		69%
DeMattia Cut Growth, 500%		9 kilocycles

DUROMETER 80 COMPOUND FOR ADHESION
Continued

Pendulum Rebound (Z): at 23°C	24%
at 100°C	39%
TG; Rheometrics (Maximum Tan Delta)	-38°C
Adhesion to RFL Tire Cord:	
Pull-Out	55 lbs./in.
Rubber Cover	100%

DUROMETER 60, TEAR STRENGTH & HEAT RESISTANCE

PPG

Neoprene GN		100.0
BR 1203		5.0
Hi-Sil 210		22.0
HAF N330		23.0
Naphthenic Oil		10.0
ODPA		2.0
Stearic Acid		1.0
Zinc Oxide		5.0
Magnesia		4.0
Total		172.0
Mooney Scorch @ 121°C; T5		30’
Viscosity, ML 100		63
Stress Strain Properties (Cure: 20’/150°C)	Original	Aged 3 D @ 100C
Hardness	62	76
M300	760(5.2)	--
T_B	2600(17.8)	2800(19.2)
E_B	720	480
Die C Tear	285(50)	340(59)
Trouser Tear (Cut Groove)	100(17.5)	55(9.6)
Compression Set 1 Day @ 100°C		73%
Yerzley Resilience		65%
Varsol Immersion 3 Days @ 23°C		
% Volume Change		40
Hardness		43

DUROMETER 70, TEAR STRENGTH & HEAT RESISTANCE
PPG

Neoprene W	100.0
BR 1203	3.0
Hi-Sil 210	26.0
HAF N326	30.0
Magnesia	4.0
ODPA	1.0
Stearic Acid	1.0
Flexricin 9	3.0
Naphthenic Oil	5.0
Zinc Oxide	5.0
Sulfur	1.0
DPG	0.5
TMTM	0.5
Ethylene Thiourea	0.2
Total	180.2
Mooney Scorch, T5	
At 121°C	14’
At 138°C	6’
Viscosity ML100	77

Stress Strain Properties (Cure: 20’/150°C)	Original	Aged 3 days @ 135°C
Hardness	70	84
M300	1270(8.6)	
Tensile	2600(17.9)	2400(16.5)
Elongation	500	280
Die C Tear	250(43.5)	260(45.4)
Trouser Tear, Cut Groove		
At 23°C	110(19.2)	40(7)
At 70°C	50(8.8)	25(4.5)
Compression Set 3 Days @ 100°C		80%

DUROMETER 65 COMPOUND FOR ABRASION RESISTANCE

PPG

Neoprene W		100.0
BR 1203		5.0
Hi-Sil 215		40.0
Mercapto Silane A189		1.0
Magnesia		4.0
ODPA		2.0
Stearic Acid		2.0
Ethylene Thiourea		0.2
Sulfur		0.2
DTDM		1.0
TMTM		0.5
DPG		0.5
Zinc Oxide		5.0
Total		161.4
Cure Rate @ 150°C, T90		38’
Mooney Scorch @ 121°C, T5		20’
Viscosity, ML100		110
Stress Strain Properties (Cure: 30’/150°C)		
	Original	Aged 3 days @ 100°C
Hardness	66	72
M300	1620(11.2)	
Tensile	3800(26.2)	3270(22.5)
Elongation	700	470
Abrasion Indices		
PCIO		119
NBS		175
Compression Set 3 Days @ 100°C		40%

PRESS MOLDED CLOSED CELL SHEET SPONGE
OIL RESISTANT MEDIUM SWELL
Uniroyal

Recipe	**1**	**2**
Neoprene W	100.0	60.0
SBR-8107	----	40.0
Stearic Acid	2.5	3.0
Magox OP	4.0	2.4
Whiting	50.0	60.0
Catalpo Clay	50.0	60.0
N550 FEF Black	30.0	20.0
Naugard® 445	2.0	2.0
Sunpar 2280	40.0	40.0
Zinc Oxide	5.0	5.0
Spider Sulfur	0.5	1.0
Diethyl Thiourea	1.0	1.0
Naugex® MBTS	12.0	12.0
Celogen® AZ-130	12.0	12.0
BIK® OT	2.0	2.0
Total	**299.0**	**309.4**
Press Precure: 15 minutes @ 121°C(250°F)		
Oven Normalize: 4 hours @ 93°C(200°F)		
Density, kg/m^3 (lb/ft^3)	192(12)	192(12)

The above compounds are typical 2B1 (RE-41-E1) types. Adjust the cure and blow systems for 2B2 (RE-42-E1) and 2B3 (RE-43-E1).

CELLULAR SHOE SOLING

Uniroyal

	1	2
SBR 8107	100.00	----
Neoprene W	----	100.00
SBR-Styrene MB	30.00	25.00
Picco 100	10.00	----
Stearic Acid	3.00	2.00
Octamine®	1.00	2.00
Magox-OP	----	4.00
Suprex Clay	55.00	50.00
Hi-Sil 233	35.00	40.00
Whiting	15.00	30.00
Zinc Oxide	5.00	5.00
Circosol 4240	10.00	18.00
Carbowax 3350	0.50	----
BIK®-OT	2.00	2.00
Retarder ESEN®	0.75	----
Diethylene GLycol	1.25	----
Naugex® MBTS	1.00	----
DPG	0.25	----
Elasto-O-Sperse PB (DETU)-75	1.25	----
Sulfur	3.00	----
OPEX 80	3.50	3.50
Total	**275.75**	**282.75**
Precure		
Time, min.	12.5	9
Temp., °C	153	168
Expansion Oven		
Time, min.	15	15
Temp., °C	153	153
Test Data		
Shrinkage 12 hours, 100°C, %	1.8	1.2
Specific Gravity	0.52	0.55
Shore A Hardness Skin	60	54
Shore A Hardness Skived	48	41